On-Track System Identification at Full Scale: Correcting and Extending a Learning-Based Pacejka Identification Pipeline for a Formula Student Vehicle in CARLA

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***Abstract*—Learning-based on-track system identification promises tire models obtained from ordinary racing laps rather than from dedicated steady-state manoeuvres. The reference method of Dikici *et al.* [1] achieves this on a 1:10 scaled platform by training a small residual neural network on measured data, rolling the corrected model out through a synthetic steering sweep, and fitting Pacejka coefficients to that synthetic sweep, iteratively. We report what happens when that pipeline is transplanted onto a full-scale Formula Student vehicle, an order of magnitude heavier and four times faster, using a CARLA-based simulation platform in which per-wheel slip angle, normal load and lateral force are directly observable. The transplant fails in a specific and instructive way: the identifier converges on a tire whose peak grip is 0.40 g, five of eight coefficients pinned on their box constraints, for a vehicle that measures 1.00–1.05 g. We show that the failure is not a solver, bounds, or tuning problem but an *excitation* problem introduced by the virtual-data step itself: the reachable friction of the synthetic rollout is $\mu_{\mathrm{reach}} = v^2\delta_{\max}/(gL)$, and a rollout speed inherited from the scaled platform caps it at 0.43 regardless of the real tire. We contribute (i) this identifiability analysis and an excitation-aware rollout that removes the pathology; (ii) an extrapolation bound on the synthetic steering sweep and a frozen-prior Tikhonov term that together stop the peak friction coefficient from drifting across co-identification iterations; (iii) three measurement-side corrections (achieved rather than commanded road-wheel angle, measured mass and geometry, and a peak-rather-than-median friction warm-start applied as a floor); and (iv) admissibility gates on derived physical quantities—axle cornering stiffness, critical speed, and grip ceiling against the raceline's lateral demand—placed at the model-to-controller boundary. After correction, three consecutive live identifications return no coefficient on a bound, peak friction coefficients of 1.05–1.18, axle cornering stiffnesses of 21 kN/rad front and 20 kN/rad rear against a simulator ground truth of 17–19 kN/rad, and one-step prediction R^2 of 0.97 for lateral velocity and 0.99 for yaw rate. We also report a negative result the corrected identifier makes visible: the optimised raceline used in this study demands 1.22 g and is therefore infeasible for this vehicle, and the admissibility gate correctly and persistently rejects deployment.**

***Index Terms*—System identification, tire modeling, Pacejka Magic Formula, vehicle dynamics, autonomous racing, model predictive control, CARLA, ROS 2, simulation-based validation, identifiability.**

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All experiments reported in this paper are performed in simulation, using the CARLA simulator and the CARLA_ASU_BRIDGE ROS 2 interface described in Section III.

I. INTRODUCTION

MODEL-BASED controllers for autonomous racing—nonlinear model predictive control (MPC), model predictive contouring control, and model- and acceleration-based pursuit—derive their performance from the fidelity of the vehicle model they carry [2], [3], [4]. In the lateral channel that fidelity is dominated by one component: the tire. The tire–road interface is the physical boundary of the vehicle's manoeuvrability, and it is also the least accessible component to measure, because its parameters depend on surface, temperature, wear and load in ways that change between sessions [5], [6].

Classical characterisation resolves this by removing the vehicle from the track: steady-state circular or ramp-steer manoeuvres in a large open space yield clean data at the cost of logistics that a race weekend rarely affords [1]. On-track identification removes the logistics but reintroduces transients, noise, and—critically—the fact that a racing line is a poor excitation signal for a nonlinear model. Nonlinear least-squares (NLS) applied directly to on-track data is known to be brittle in exactly this regime [1].

Dikici *et al.* [1] propose a hybrid remedy that is the direct baseline for this paper. A small residual multilayer perceptron is trained on 30 s of ordinary lap data to predict the one-step error of a nominal single-track model. The corrected model—nominal plus residual—is then rolled out through a *synthetic*, quasi-steady steering sweep, and Pacejka coefficients are fitted to that synthetic sweep by least squares. Because the sweep is generated rather than driven, it can be made steady and noise-free; because the residual network only has to interpolate within the distribution it was trained on, its known generalisation weakness is contained. The procedure is applied iteratively, each fit becoming the next nominal model. On a 1:10 scaled platform the method reports a 3.3× lower one-step RMSE than NLS under noise and tire models comparable to steady-state identification, from 30 s of driving and 3 s of training [1].

A. Problem statement

This paper asks what happens when that pipeline is moved from a 3.7 kg, 0.32 m-wheelbase RC platform to a full-scale Formula Student vehicle: 269.6 kg, 1.53 m wheelbase, 16 ° of road-wheel lock, driven at 11–27 m/s. The move is not a matter of editing a parameter file. Two scale-dependent mechanisms, both invisible at 1:10, break the identification:

1) **Excitation collapses in the virtual-data step.** The Pacejka fit never sees the recorded lap; it sees the synthetic rollout. The friction that rollout can reach is bounded by $\mu_{\mathrm{reach}} \approx v^2\delta_{\max}/(gL)$, which is a property of the *rollout configuration*, not of the tire. A rollout speed inherited from the scaled platform caps μ_{reach} at 0.43 on this vehicle. Below the tire's peak, the Magic Formula degenerates to $F_y \approx F_z(BCD)\alpha$: only the product BCD is identifiable, and the optimiser parks the individual coefficients wherever the box constraints stop it [7], [5].
2) **Lateral demand scales as v^2, so a bad grip estimate is silent until it is catastrophic.** An identified peak friction coefficient of 0.40 produces a 0.03 m cross-track error at 11 m/s and 41.89 m at 27 m/s in our closed-loop harness. The scaled platform never reaches the speed at which the error becomes observable.

A third difference is an opportunity rather than an obstacle. Because the experiments are conducted in the CARLA simulator [8] through a custom ROS 2 interface, the simulator's own per-wheel telemetry—slip angle, normal load, lateral force, per wheel—is available as ground truth. An on-track identifier can therefore be validated against the forces the plant actually generated, which is not possible on a physical car without a force-measuring wheel hub.

B. Contributions

1) **An identifiability analysis of the virtual-data step**, with the closed-form reachable-friction bound above, together with an *excitation-aware rollout* that selects the rollout speed from the measured lap and warns whenever the configuration cannot reach the prior's peak. This is a correction to the baseline method [1]—applicable at any scale, but latent at 1:10 because a 2–4 m/s rollout is representative there and is not here (Sections III-G1, III-G).
2) **Two mechanisms that stabilise the iterative co-identification.** We bound the synthetic steering sweep at $1.5\times$ the 99th percentile of the steering actually observed, so the fit cannot read the residual network's extrapolation as grip; and we freeze the Tikhonov regularisation target at the configured prior while letting the optimiser's start point track the iterations. Together these remove a 1.01→1.87 drift of the peak friction coefficient over six iterations (Section III-H).
3) **Measurement-side corrections** for full-scale deployment: slip angles built from the *achieved* road-wheel angle rather than the command (measured ratio 0.76, worth 20 % of front axle stiffness); mass and wheelbase taken from measured static wheel loads rather than from the configuration file; and a friction warm-start computed as a high quantile over the whole buffer and applied as a *floor* on the prior, rather than as a median over an explicitly low-slip mask (Section III-J).
4) **Admissibility gates on derived physical quantities** at the model-to-controller boundary—axle cornering stiffness, understeer sign and critical speed, and grip ceiling against the raceline's peak lateral demand—rather than per-coefficient box checks, which cannot see a degenerate fit or a pairwise-oversteering axle set (Section III-K).
5) **A simulation platform for on-track identification research** (CARLA_ASU_BRIDGE) that exposes the plant's own per-wheel forces, slip angles and loads as ROS 2 topics alongside a conventional autonomy interface, enabling the ground-truth comparisons reported here (Section III-C).
6) **An honest negative result.** With the identifier corrected, the model reports a grip ceiling of 1.02–1.16 g against a measured steady-state ceiling of $\approx 1.0\,g$, while the optimised raceline used in this study demands $1.22\,g$. The admissibility gate therefore refuses the identification for closed-loop use—correctly. We report this rather than loosening the gate, and quantify the raceline regeneration it implies (Section IV-F).

All results in this paper are obtained in simulation. Section V states explicitly which conclusions are expected to transfer to a physical vehicle and which are properties of the simulator's tire model.

II. Related Work

A. Tire modeling and Magic Formula identification

The Magic Formula of Bakker, Nyborg and Pacejka [9] and its subsequent consolidation [10], [5] remains the standard empirical description of steady-state tire force generation, and is the model identified in this work. Its lateral form,

$$F_y = F_z\, D \sin\!\Big(C \arctan\!\big(B\alpha - E(B\alpha - \arctan B\alpha)\big)\Big), \quad (1)$$

is parameterised by a stiffness factor B, shape factor C, peak factor D and curvature factor E. Two properties of (1) are structural and are used repeatedly in this paper. First, its slope at zero slip is F_zBCD, so the axle cornering stiffness depends on the coefficients only through that product and E drops out. Second—and this is the identifiability statement—if the data does not approach the peak, (1) linearises to $F_y \approx F_z(BCD)\alpha$ and B, C, D become mutually unidentifiable. This is a standard persistent-excitation argument [7]; we do not claim it as novel, but we do claim that the baseline pipeline's virtual-data step can silently violate it, and that the violation is what produced the failure we report.

Classical characterisation obtains the coefficients from steady-state manoeuvres in a controlled space [1], [6], which yields clean data at high logistical cost. Direct fitting to measured contact-patch forces, where a force-measuring hub or a simulator's own wheel telemetry is available, is the most informative variant; in this work it is used only as a *reference* against which the on-track identifier is scored, never as part of the identification loop.

B. On-track and data-driven identification

On-track methods trade excitation quality for operational simplicity. NLS applied to lap data is the classical baseline and is known to be brittle under noise [1]. Gaussian-process approaches learn the model mismatch of a nominal model and have been used both for identification and directly inside the controller [11], [12], [3]; they are accurate but carry a continuing computational cost and, as [1] observes, are usually evaluated starting from an already accurate physical model. Neural approaches replace the GP with a cheaper regressor at the cost of generalisation guarantees.

Deep Dynamics [13] is the closest work on the physics-constraint axis: it estimates Pacejka coefficients inside a neural network and adds a *Physics Guard* layer that clamps the internal coefficient estimates into nominal physical ranges. Our admissibility gates share that motivation but differ in placement and in what is constrained. We found—and report as a substantive finding—that per-coefficient box constraints are *insufficient*: the degenerate fit that motivated this work satisfied every box constraint while five of eight coefficients sat exactly on a bound, and a later fit satisfied every box constraint while being pairwise oversteering and open-loop unstable above 17.4 m/s. We therefore gate on derived quantities (cornering stiffness, understeer sign, critical speed, grip ceiling versus the trajectory's demand) at the model-to-controller interface, and treat a coefficient sitting on a bound as a diagnostic of non-identifiability rather than as a successful clamp.

The hybrid residual-plus-virtual-data structure of Dikici *et al.* [1] is the baseline we extend, and is described in detail in Section III. Its key idea—use the learned model only where it interpolates, and extract physically parameterised coefficients from it—is retained unchanged. What we correct is the mechanism that decides *where* the learned model is queried.

C. Overlap with concurrent work

Wu *et al.* [14] extend the same baseline pipeline concurrently and independently, and the overlap must be stated plainly. They introduce (i) a vision-based friction prior from a lightweight CNN over road texture, used to warm-start the optimisation; (ii) an S4 structured state-space model in place of the MLP, to capture temporal residuals; and (iii) Nelder–Mead for the iterative Pacejka extraction, with virtual simulation in CarSim. The pipeline described in this paper independently contains an S4-family temporal residual architecture (an S4D variant [15], [16] evaluated by recurrent scan rather than FFT convolution, selected for the short $\approx$1500-sample sequences here), a friction warm-start, and Nelder–Mead and differential-evolution solver options alongside trust-region reflective.

We therefore do *not* claim the temporal-residual architecture, the friction warm-start concept, or the derivative-free solver as contributions of this work. Two differences are worth recording because they are technical rather than merely chronological. First, our friction prior is proprioceptive—derived from measured accelerations, in the spirit of model-free friction estimation from IMU data [17]—rather than visual, and therefore does not require a camera or a textured-surface assumption. Second, and more importantly, we find that a friction prior computed as *utilised* friction is a lower bound on *available* friction and cannot substitute for excitation: on a gentle lap the median utilisation on our vehicle measures 0.044. We consequently apply the prior as a floor rather than as a replacement, and locate the actual fix in the rollout excitation rather than in the initialisation. To our knowledge the reachable-friction bound of Section III-G and its consequence for the virtual-data step have not previously been stated.

D. Learning-based and model predictive control for racing

The downstream consumer of the identified model in this work is a nonlinear MPC built on `acados` [18], using HPIPM partial condensing [19] and, in earlier configurations, qpOASES [20]. The path-tracking formulation follows standard practice in racing MPC [2], [3]; the reference raceline is generated by minimum-curvature optimisation in the manner of Heilmeier *et al.* [21]. Our contribution on this side is not a new controller but the observation, made concrete in Section III-K, that an identified tire set can be numerically inadmissible for the controller in ways the identifier itself has no way to see—stiff linearisation, oversteering axle pairs, and grip ceilings below the trajectory's demand—and that these are best enforced as an explicit contract at the interface.

E. Simulation-based validation and CARLA

CARLA [8] is the standard open simulator for autonomous-driving research and provides the sensor and scenario infrastructure used here. Its native vehicle dynamics come from the underlying PhysX wheeled-vehicle model, which is a rigid-body model with per-wheel tire friction rather than an empirical Magic Formula fit to measured data; this is the reason our study treats the simulator as a well-instrumented *plant* to be identified, not as a source of ground-truth Pacejka coefficients. Recent work has addressed exactly this fidelity gap: R-CARLA [22] makes the dynamics model interchangeable while retaining CARLA's sensor simulation and reports substantial sim-to-real gap reductions, and Sagmeister *et al.* [23] quantify how far a vehicle model can be simplified before conclusions about a trajectory-following controller stop transferring—finding, relevantly for us, that the sensitivity is largest near the acceleration limits. CARLA has also been used as the validation environment for complete Formula Student Driverless stacks with ROS 2 interfaces [24], and for digital-twin racing pipelines. The platform described in Section III-C sits in this line: it is a ROS 2 bridge around a Formula Student vehicle model in CARLA, distinguished by exposing the simulator's own per-wheel slip, load and force telemetry so that an identification method can be scored against the forces the plant actually produced.

Scaled autonomous racing platforms—F1TENTH [25] and the ForzaETH Race Stack [4]—provide the software context of the baseline. Full-scale Formula Student Driverless systems [6] provide the vehicle class targeted here.

F. Gap addressed

Summarising: the baseline [1] establishes that a residual network plus virtual steady-state data can recover a Pacejka model from a lap, at 1:10 scale. Concurrent work [14] improves its residual architecture and its initialisation. Neither

Loop diagram of the four identification stages, with this paper's four changes highlighted.

Fig. 1. The identification pipeline. Recorded lap data trains the residual network; the corrected model is rolled out through a synthetic steering sweep; Pacejka coefficients are fitted to that sweep; the fit becomes the next nominal model. Highlighted: the excitation-aware rollout speed (Section III-G), the extrapolation-bounded sweep and frozen regularisation target (Section III-H), and the admissibility gate on the output (Section III-K).

addresses the condition under which the virtual data itself carries information about the tire's peak, which is the binding constraint at full scale. Neither treats the admissibility of the identified model for the controller that will consume it as an explicit, physically derived contract. This paper addresses both, and does so on a platform where the answer can be checked against the plant's own forces.

III. Method

Figure 1 shows the complete pipeline. Subsections III-A–III-B restate the baseline [1] to fix notation; III-C–III-D describe the experimental platform; and III-G–III-K describe this paper's changes. Section A lists the assumptions made where an implementation detail could not be measured.

A. Vehicle model

We use the dynamic single-track (bicycle) model of the baseline, with rigid body mass m, yaw inertia I_z, and centre-of-gravity distances l_f, l_r to the front and rear axles, $L = l_f + l_r$. Longitudinal and lateral dynamics are decoupled and load transfer is neglected:

$$\dot{v}_y = \tfrac{1}{m}\big(F_{y,r} + F_{y,f}\cos\delta - m v_x \omega\big), \tag{2}$$

$$\dot{\omega} = \tfrac{1}{I_z}\big(F_{y,f} l_f \cos\delta - F_{y,r} l_r\big). \tag{3}$$

Slip angles follow from the kinematic relations

$$\alpha_f = \delta - \arctan\!\Big(\tfrac{v_y + l_f\omega}{v_x}\Big), \quad \alpha_r = -\arctan\!\Big(\tfrac{v_y - l_r\omega}{v_x}\Big), \tag{4}$$

and the axle forces from the Magic Formula (1) with per-axle coefficients $\Phi_p = [B_f, C_f, D_f, E_f, B_r, C_r, D_r, E_r]$. The identification state and input are $\mathbf{x} = [v_y, \omega]$ and $\mathbf{u} = [v_x, \delta]$.

The baseline discretises (2)–(3) by explicit Euler at T_s. We retain Euler as the default but expose second- and fourth-order Runge–Kutta as configuration options (`ode_solver`), because the full-scale vehicle's lateral mode is stiffer relative to T_s than the scaled platform's; the same concern appears in a sharper form in the downstream MPC (Section III-K). The nominal one-step map is written $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$.

Two derived quantities are used throughout and are worth naming explicitly. The linearised axle cornering stiffness is the slope of (1) at zero slip,

$$C_i = F_{z,i} B_i C_i^{\mathrm{MF}} D_i, \qquad i \in \{f, r\}, \tag{5}$$

(writing C^{MF} for the Magic Formula shape factor to avoid a clash), and the grip ceiling implied by an identified set is

$$a_y^{\max} = \frac{D_f F_{z,f} + D_r F_{z,r}}{m}. \tag{6}$$

The linear single-track model built from (5) is oversteering when $l_f C_f > l_r C_r$ and is then open-loop unstable above the critical speed $v_{\mathrm{crit}} = L\sqrt{C_f C_r / \big(m(l_f C_f - l_r C_r)\big)}$ [26].

B. Baseline identification pipeline

The baseline proceeds in four stages, repeated for N_{it} iterations [1]:

1) **Residual learning.** A small MLP f_{NN} is trained on the recorded lap to predict the one-step error $e_k = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}$ from $[v_x, v_y, \omega, \delta]_k$. The corrected model is $\mathbf{x}_{k+1} = \hat{\mathbf{x}}_{k+1} + e_k$.
2) **Virtual steady-state generation.** The corrected model is integrated open-loop at a constant longitudinal speed while the steering angle is ramped linearly from 0 to δ_{sweep}, producing a synthetic quasi-steady dataset.
3) **Pacejka fit.** Assuming $\dot{v}_y = \dot{\omega} = 0$ on the synthetic data, axle forces are recovered from the yaw balance,
$$F_{y,r} = \frac{m l_f v_x \omega}{L}, \qquad F_{y,f} = \frac{m l_r v_x \omega}{L\cos\delta}, \tag{7}$$
slip angles from (4), and Φ_p is fitted to $(\alpha_i, F_{y,i})$ by bounded least squares.
4) **Re-nomination.** The fitted Φ_p becomes the nominal model of the next iteration and the residual network is reinitialised.

The crucial structural point, and the one that motivates Section III-G, is that stage 3 *never sees the recorded lap*. It sees the output of stage 2. The recorded lap enters only through the residual network.

C. Experimental platform: Formula AI vehicle in CARLA

The plant is a Formula Student–class driverless vehicle model, `vehicle.vehicle.asurt_fsai`, instantiated in CARLA [8] (server build 0.9.16, Unreal Engine 4.26) on the `silverstone` map. It is a four-wheeled, front-steered, all-wheel-drive rigid-body vehicle simulated by the engine's PhysX wheeled-vehicle solver; lateral force generation is produced by the solver's per-wheel tire model under a configured friction coefficient, not by an analytic Magic Formula. This distinction matters for interpreting the results: the simulator has no "true" Pacejka coefficients to recover, so the reference tire curve used in Section IV is itself a Magic Formula fit to the simulator's own per-wheel force/slip/load telemetry, obtained under a dedicated limit-excitation manoeuvre. The vehicle is shown in Fig. 2.

Table I contrasts configured against measured properties, and three rows in it are load-bearing for the rest of the paper.

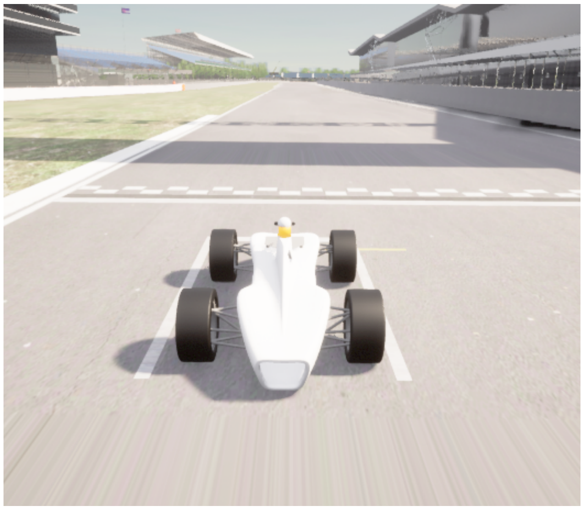

Fig. 2. The Formula AI vehicle model, `vehicle.vehicle.asurt_fsai`, on the start/finish straight of the `silverstone` map in CARLA. This open-wheel Formula Student–class vehicle is the plant identified throughout this paper; its lateral forces are produced by the engine's PhysX wheeled-vehicle solver rather than by an analytic tire model, which is why the reference curve of Table III is fitted to the simulator's own per-wheel telemetry. Measured geometry and mass are given in Table I ($l_f = 0.738$ m, $l_r = 0.795$ m, $L = 1.533$ m, wheel radius 0.25 m, $\pm 16°$ road-wheel limit, 269.6 kg). The IMU and GNSS sensors are mounted at the vehicle origin (zero offset in the bridge configuration).

TABLE I

VEHICLE CONFIGURATION AND MEASURED PLANT PROPERTIES. CONFIGURED VALUES ARE SET IN THE BRIDGE'S `CARLA_INTERFACE_CONFIG.YAML`; MEASURED VALUES ARE OBTAINED FROM THE SIMULATOR'S PER-WHEEL TELEMETRY.

Quantity	Value	Unit
Mass m	240.0	kg
Static wheel load (total)	2644.5	N
Front/rear static split	51.2 / 48.8	%
Wheelbase L	1.5334	m
l_f / l_r	0.7381 / 0.7953	m
Yaw inertia I_z	51.1	kg m^2
Wheel radius	0.25	m
Max road-wheel angle	16.0	°
PhysX `tire_friction`	1.5	—
Realised peak axle μ	**1.00–1.05**	—
Peak transient a_y	1.50	*g*
Steady-state a_y ceiling	≈1.0	*g*
Axle cornering stiffness	$\approx 12.1\, F_z$	N/rad
front / rear	17198 / 16516	N/rad
Drag coefficient	0.426	—

[a]Assumed; see Section A.

First, the mass the physics engine actually simulates (269.6 kg, measured as total static normal load at rest) is 12 % above the configured 240 kg; since (7) is linear in m, using the configured value biases every "measured" force by the same 12 %. Second, the PhysX wheel material friction coefficient (1.5) is *not* the realised peak axle friction (1.00–1.05); an identification that converges toward 1.5 because the configuration file says so is converging on the wrong target. Third, the achieved road-wheel angle is 0.76 of the commanded angle, so slip angles built from the command are systematically too large.

TABLE II

ROS 2 INTERFACE USED BY THE IDENTIFICATION AND VALIDATION PIPELINE. NAMESPACE IS `/SIM` UNLESS MARKED ABSOLUTE.

Topic	Type	Role
`/odom`	`nav_msgs/Odometry`	State v_x, v_y, ω; twist is body-frame
`feedback/steering_angle`	`std_msgs/Float32` (deg)	Achieved road-wheel angle δ
`feedback/imu`	`sensor_msgs/Imu`	a_x, a_y for the friction warm-start
`feedback/tire_forces`	`sim_manager_msgs/TireForces`	Per-wheel α, F_z, F_y: **ground truth**
`feedback/speed`	`std_msgs/Float32`	Forward speed echo
`feedback/motors`	`std_msgs/String` (JSON)	Per-wheel speed/torque/brake
`/drive`	`ackermann_msgs/AckermannDriveStamped`	Control command in
`control/tire_friction`	`std_msgs/Float32`	Runtime grip override
`/clock` *(abs.)*	`rosgraph_msgs/Clock`	Simulation time base

D. CARLA_ASU_BRIDGE: *ROS 2 interface and telemetry*

CARLA_ASU_BRIDGE is a C++ ROS 2 lifecycle node (`carla_telemetry_cpp`) linked directly against CARLA's C++ client library. It spawns and configures the ego vehicle, drives its control inputs, and republishes the simulator's state as standard ROS 2 topics. All identification traffic in this paper crosses this interface. Figure 3 should render the resulting architecture.

1) Command path: The bridge is configured in `ackermann_drive` mode: it subscribes to a single `ackermann_msgs/AckermannDriveStamped` on `/drive` and forwards speed and steering angle to CARLA's native Ackermann controller, whose server-side cascaded speed/acceleration PID loops are configured once at startup. No client-side steering PID is present—the commanded road-wheel angle is passed through in radians—which is why the achieved/commanded ratio of Table I is a property of the vehicle and its steering curve rather than of a tuning choice in the bridge. A speed cap (`max_velocity_kmh`) is applied in both control modes.

2) Feedback path used for identification: Table II lists the interface actually consumed by the identifier and its validation. Three properties of this interface are specific to this platform and are what make the study possible.

a) Per-wheel ground truth: `feedback/tire_forces` is a direct passthrough of CARLA's own `WheelTelemetryData`: per-wheel lateral slip angle, normal load, lateral force, longitudinal force and wheel torque, in wheel order [FL, FR, RL, RR]. There is no analytic tire model inside the bridge, so nothing on this topic can disagree with the physics the vehicle is being driven by. The lateral force is sign-flipped and the slip angle converted from degrees into the ROS right-handed body frame ($+y$ left); the longitudinal force is *not* a contact-patch force—CARLA computes it as $-\tau/r_w$, i.e. drivetrain torque at the axle—and is therefore not used in this work. This topic is what allows an on-track identifier to be scored against the forces the plant actually generated.

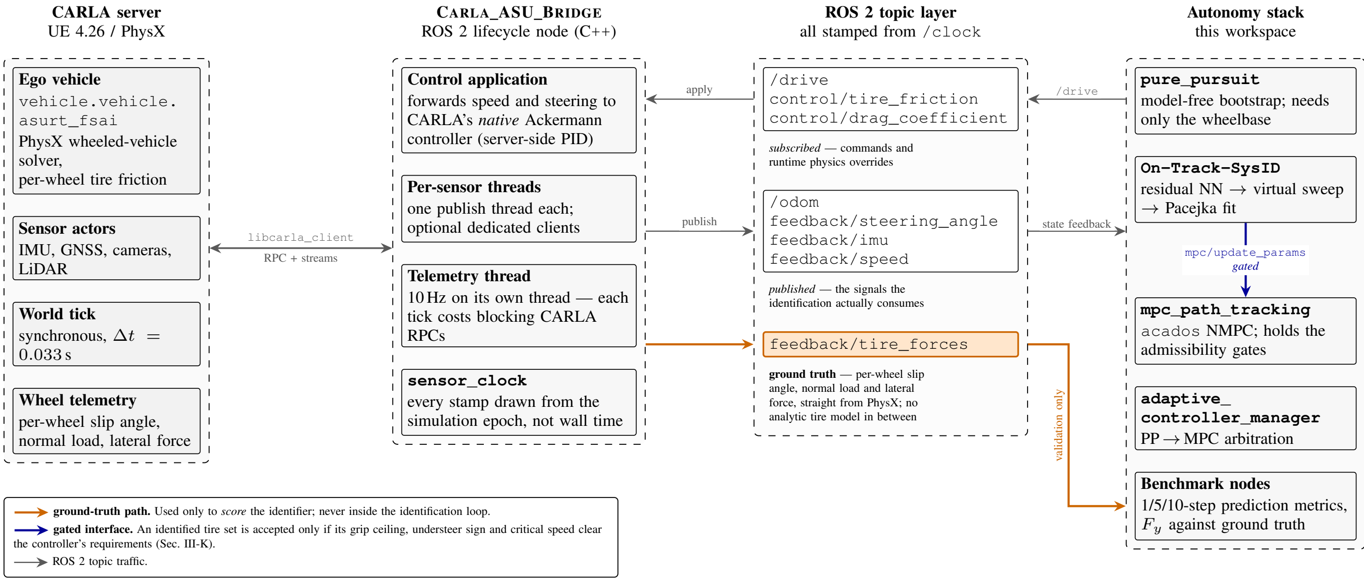

Fig. 3. Architecture of the CARLA_ASU_BRIDGE ROS 2 interface. The bridge links against CARLA's C++ client, drives the ego vehicle through CARLA's native Ackermann controller, and republishes state, sensors and per-wheel wheel telemetry on ROS 2 topics stamped from the simulation clock. The `feedback/tire_forces` path (highlighted) carries the simulator's own per-wheel slip angle, normal load and lateral force, and is the ground truth against which the identifier is scored.

b) Simulation-time stamping: Every feedback message is stamped from the same simulation-time epoch as `/clock`, not from wall-clock time. In synchronous mode CARLA ticks as fast as the server manages (observed near $1.9\times$ real time on our hardware), so wall-clock stamping would drift the identification dataset against itself without bound.

c) Odometry and QoS: Odometry is computed from the actor transform and velocity, published at 50 Hz with `follow_server_rate` enabled, with an optional additive Gaussian noise model (disabled for the runs reported here). The publisher offers `BEST_EFFORT`; every subscriber in the identification stack requests `BEST_EFFORT` accordingly. We record this because a reliability mismatch on this interface is silent—DDS requires requested $\leq$ offered, so a `RELIABLE` request against a `BEST_EFFORT` publisher produces one warning and then no data at all.

d) Verification of frame conventions: Because (4) is meaningless if v_y is expressed in the wrong frame, the body-frame assumption on `/odom`'s twist was verified directly rather than assumed: correlation between `twist.linear.y` and the body-frame lateral velocity reconstructed from pose and world velocity is 0.93 with 0.09 m/s RMS error, against 16.7 m/s RMS under a world-frame interpretation.

3) Modifications for identification: The following changes were made to the vehicle, sensing or control stack specifically to support this study, and are listed so that the setup is reproducible:

- *Steering feedback exposed to the identifier.* `feedback/steering_angle` existed but had no subscribers; the identifier now consumes it (with configurable units, scale and staleness timeout) and falls back to the `/drive` setpoint when it is silent.
- *Runtime grip and drag overrides.* `control/tire_friction` and `control/drag_coefficient` write straight to the live `VehiclePhysicsControl`, allowing the surface friction to be changed within a run without respawning—the intended mechanism for future surface-change experiments (Section VI).
- *Drivetrain and tire tuning for low-speed stability.* The torque curve, zero-throttle damping rates and wheel `tire_friction` were tuned against a low-speed hold test; a wheel friction coefficient above $\approx$2.0 on a steered wheel scrubs hard enough at creep speed to stall the drivetrain intermittently. These values define the plant being identified and must be reported with any tire result.
- *Telemetry on a dedicated thread* at 10 Hz, off the control loop and the ROS executor, because each telemetry tick costs several blocking CARLA RPCs.

E. Data collection and preprocessing

The vehicle is driven around the raceline by a model-free pure-pursuit controller, which requires no tire knowledge—only the wheelbase—and is therefore a valid bootstrap, exactly as in the baseline [1], [4]. Odometry, the achieved steering angle and the drive command are sampled into a ring buffer at 50 Hz ($T_s = 0.02$ s) for 30 s, gated on $v_x > 1$ m/s. Preprocessing follows the baseline: a zero-phase low-pass filter, then symmetry augmentation by mirroring $(v_y, \omega, \delta) \rightarrow (-v_y, -\omega, -\delta)$ with v_x unchanged, which both doubles the data and removes the cornering bias of a circuit. Re-identification is retriggered every `reidentification_interval_s` (30 s by default).

F. Residual model

The default residual architecture is the baseline MLP: 4 inputs, one hidden layer of 8 units, 2 outputs, LeakyReLU, 58 parameters, Adam at 5×10^{-4}, full-batch, MSE. Because the full-scale vehicle's residual has different structure from the scaled platform's, the architecture is now selectable (`nn_architecture`): `baseline`, `wide`, `physics_inputs` (slip angles appended to the input vector), `ensemble`, and `s4`—an S4D diagonal state-space temporal residual [15], [16] evaluated by causal recurrent scan over a sliding window of 20 samples (0.4 s, matched to tire-relaxation timescales). As noted in Section II, the S4-family residual overlaps with concurrent work [14] and is not claimed as a contribution.

An optional physics-informed loss [27], [13] is available as an ablation switch (`loss_mode`), adding to the data term a dynamics-consistency residual evaluated on the corrected one-step prediction, a left/right symmetry term, and an input-gradient smoothness term, with configurable weights. The default configuration for the results reported here is `loss_mode: mse`, i.e. the baseline objective; the physics-informed variant is reported as an ablation (Section IV-E).

G. Excitation-aware virtual rollout

This is the central correction. Consider stage 2 of Section III-B: a constant-speed rollout at v with the steering ramped to δ_{sweep}. In quasi-steady cornering the lateral acceleration a single-track vehicle can develop kinematically is $a_y = v^2\kappa$ with $\kappa \approx \delta/L$, so the largest friction coefficient the rollout can *demand* of the tire is

$$\boxed{\mu_{\text{reach}} \;\approx\; \frac{v^2\,\delta_{\text{sweep}}}{g\,L}} \tag{8}$$

Equation (8) contains no tire parameter. It is a property of the rollout configuration alone. If $\mu_{\text{reach}} < D$, the synthetic data never leaves the linear ramp of (1), only the product BCD is identifiable [7], and the bounded optimiser resolves the remaining freedom by sliding onto a box edge—which is precisely the signature we observed.

The baseline implementation selects the rollout speed as $v = \text{clip}(\overline{v_x},\, 2,\, 4)$ m/s, a value representative of a 1:10 platform. On our vehicle, with $\delta_{\text{sweep}} = 0.4$ rad and $L = 1.5334$ m, (8) gives $\mu_{\text{reach}} = 0.43$ at 4 m/s: the rollout cannot ask the tire for more than 0.43 g no matter what the tire is. At 1:10 scale, with $L \approx 0.32$ m, the same rollout speed yields $\mu_{\text{reach}} \approx 2.0$ and the pathology is invisible.

1) Correction: The rollout speed is now taken from the speed the vehicle was actually driven at, clamped by configuration bounds (`pacejka_rollout.speed_min=` 7 m/s, `speed_max=null`), and the pipeline emits a warning whenever (8) evaluates below the prior's D—turning a silent degeneracy into a startup diagnostic. The bound in (8) is also the right tool for choosing a data-collection speed: identification data should be gathered at $v \geq \sqrt{gLD/\delta_{\text{sweep}}}$.

H. Bounded extrapolation and frozen-prior regularisation

Raising the rollout speed exposes a second, opposite failure. The residual network is queried by the rollout at steering angles the network has never seen: on a pure-pursuit lap the observed $|\delta|$ stays below ≈ 0.06 rad while the sweep ran to 0.4 rad. The fit then reads the network's extrapolation as additional grip, and because each iteration's answer became both the next iteration's nominal model *and* its regularisation target, the error compounded: D walked from the 1.009/1.002 prior to 1.5402/1.8721 over six iterations, i.e. a claimed 1.7 g on a vehicle measuring 1.0 g in steady state.

Two changes close this loop:

1) **Extrapolation bound.** The sweep terminates at $\delta_{\text{sweep}} = \text{clip}\big(\lambda_e \cdot P_{99}(|\delta_{\text{obs}}|),\, \delta_{\min},\, \delta_{\max}\big)$ with $\lambda_e = 1.5$, $\delta_{\min} = 0.05$ rad, $\delta_{\max} = 0.34$ rad. The network is never queried more than 50 % beyond the steering range it was trained on.
2) **Frozen prior.** The Tikhonov target of the fit is frozen at the configured prior Φ_p^0 for all iterations, while the optimiser's *start point* continues to track the previous iteration. The objective per axle is

$$\min_{\phi} \sum_k \rho\big(r_k(\phi)\big) \;+\; \sum_j w_j \big(\phi_j - \phi_j^0\big)^2, \tag{9}$$

with r_k the force residual, ρ a soft-ℓ_1 robust loss, and w_j scaled to the data residual budget by a single dimensionless weight $\lambda = 0.05$. The purpose of the regulariser is specifically to hold the coefficients the manoeuvre does not excite—E above all—at their nominal value instead of on a box edge.

I. Pacejka fit

Fitting uses the steady-state force recovery of (7) on the synthetic data, with box constraints $\Phi_p \in ([4.0, 1.2, 0.4, -3.0], [20.0, 2.2, 2.0, 1.0])$ per axle. Relative to the baseline we (i) raised the number of jittered multi-starts from 1 to 8, (ii) added a derivative-free simplex (Nelder–Mead) and a global differential-evolution solver as selectable alternatives to trust-region reflective, and (iii) corrected the configured prior, which previously initialised D *below* its own lower bound and thus started every optimisation on the rail it then failed to leave. The prior now carries the measured tire of Table III, fitted directly to the simulator's per-wheel telemetry.

We stress that none of (i)–(iii) fixes the degeneracy on its own. They were verified as secondary: with the rollout speed left at 4 m/s, a synthetic $D = 1.5$ tire fits back as $D_f = 0.753$, $D_r = 0.737$ with three coefficients railed, regardless of solver or start; at 8–20 m/s the same tire fits back as $D = 1.40$–1.61 with nothing railed.

J. Measurement-side corrections

Three corrections apply to the measurement chain rather than to the estimator.

1) Achieved versus commanded steering: α_f in (4) was built from the `/drive` setpoint. The measured achieved/commanded ratio on this vehicle is 0.76, so α_f was systematically $\approx$ 25 % too large and the fitted front stiffness $\approx$ 20 % too small (13725 N/rad against the simulator's 17198 N/rad on the same run). Using `feedback/steering_angle` recovers 18486 N/rad.

2) Mass and geometry: m and l_{wb} are taken from the measured static wheel loads (Table I) rather than from the configuration file, and $l_{wb} \equiv l_f + l_r$ is enforced by a regression test, because (7) uses the wheelbase for the normal loads and the axle split for the force division; the two disagreeing is a silent bias.

3) Friction warm-start as a floor: An initial estimate of D is derived from measured accelerations, in the spirit of model-free friction estimation [17]. The baseline-adjacent implementation computed the *median* of $\sqrt{a_x^2 + a_y^2}/g$ over an explicitly *low-slip* mask. This is structurally wrong: it measures utilised friction over samples selected for being far from the limit, and utilisation is a lower bound on capability. On a 13 m/s pure-pursuit lap the median measures 0.044 while the peak measures 0.46. The estimator is now a 99th-percentile over the whole buffer and is applied as a *floor* on the prior's D, never as a replacement, and only on the first identification of a node lifetime. We report this as a corrected implementation detail, not as a criticism of the underlying friction-estimation literature, whose methods target *available* friction and assume near-limit excitation.

K. Deployment: admissibility gates

The identified Φ_p is pushed to a nonlinear MPC path-tracking controller (`acados` [18], $N = 100$, $dt = 0.02$ s raised to the 50 Hz control period, partial-condensing HPIPM [19]) over a ROS 2 service. We found that a coefficient set can be simultaneously inside every box constraint and unusable, in three distinct ways, and that each needs a gate on a *derived* quantity:

1) **Grip ceiling versus trajectory demand.** The controller computes $\max_k(v_{x,k}^2|\kappa_k|)$ over the speed-clamped reference and rejects any identification whose $a_y^{\max}$ from (6) falls below it. The degenerate fit reported a $0.40\,g$ ceiling against a $1.22\,g$ demand and produced 41.89 m of cross-track error at 27 m/s in closed-loop simulation, against 0.03 m for a physically plausible set.
2) **Understeer sign and critical speed.** A set with $l_f C_f > l_r C_r$ whose v_{crit} falls inside the controller's speed range makes the model open-loop unstable; the resulting $\rho(A_d) > 1$ is raised to ρ^N in the condensed QP and the solve fails. An observed set gave $C_f = 37031$ N/rad against $C_r = 11691$ N/rad, i.e. $v_{\text{crit}} = 17.4$ m/s.
3) **Integration stiffness.** Even a stable set can be stiff relative to the control period. A single explicit RK4 step at $dt = 0.05$ s gave $\rho(A_d) = 547$ at 5 m/s for one identified set. The model now sub-steps adaptively, sizing the sub-step from the model's own fastest lateral mode $|\lambda| \approx (C_f + C_r)/(mv_x) + (l_f^2 C_f + l_r^2 C_r)/(I_z v_x)$ so that $|\lambda| h \le 1.5$, which restores $\rho \approx 1.000$ across the speed range at a measured cost of 2.3–6.8 ms per 100-stage horizon.

Two further controller-side changes were required before the identified model could be evaluated in closed loop at full-scale speeds, and are reported for completeness rather than as identification contributions: reference speeds are clamped at ingest (the raceline's 14.9–46.0 m/s against a plant cap silently corrupted the speed target, the yaw-rate reference, the horizon arc-length advance and the adaptive step size at once), and the solved state is rolled forward by the measured transport delay (odometry age + control period + last solve time) before each solve, which took cross-track error at 27 m/s from 130.55 m to 0.38 m.

L. Validation methodology

Three levels of validation are used, in increasing strength:

1) **One-step prediction** on a held-out run, as in the baseline: RMSE and R^2 of v_y and ω under $\hat{\mathbf{x}}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k, \Phi_p)$. A dedicated ROS 2 node extends this to configurable multi-step horizons (1/5/10 steps) with Euler/Heun/RK4 integration and online Welford statistics.
2) **Direct force comparison** against `feedback/tire_forces`: the identified curve is evaluated at the measured (α, F_z) and compared to the measured F_y, per axle. This is the level unavailable on a physical platform and is the strongest evidence in the paper.
3) **Closed-loop tracking** under the MPC using the identified model: lap time, mean and maximum lateral deviation, steering sign reversals per control step, and RMS steering rate against the rate limit. This level is *not* completed in this paper; see Section IV-F.

The steady-state force recovery of (7) was itself validated against the simulator's own axle force on a normal lap: correlation 0.993, slope 0.94. This matters because (7) is not a force measurement—it is fully determined by $v_x\omega$—and its adequacy had to be established rather than assumed.

IV. Experiments and Results

A. Experimental setup

All experiments are conducted in simulation. The plant is the Formula AI vehicle of Section III-C in CARLA [8], driven through CARLA_ASU_BRIDGE on a ROS 2 Humble workspace; the CARLA server runs in synchronous mode at a 0.033 s fixed physics step. Identification data is collected over 30 s of pure-pursuit driving at 50 Hz, matching the baseline's data budget [1]. Ground truth for tire forces, slip angles and normal loads is the simulator's own per-wheel telemetry.

1) Reference trajectory: Every run follows the same optimised raceline, shown in Fig. 4. It was produced with the TUM `global_racetrajectory_optimization` package [28], vendored in the simulator workspace, using its minimum-curvature formulation [21] on a recorded centreline of the CARLA `silverstone` circuit. The result is a closed 5830 m lap of 2916 waypoints at 2.0 m spacing, with a reference speed of 14.90–46.02 m/s and a minimum corner radius of 18.7 m ($|\kappa|_{\max} = 0.0536$ /m).

Two of the optimiser's inputs matter for Section IV-F and are therefore stated explicitly. Its vehicle configuration assumes a mass of 204 kg, against the 269.6 kg the physics engine actually simulates (Table I). More consequentially, its `ggv` envelope declares a *constant* lateral capability of 12.0 m/s^2 at every speed. The resulting profile saturates that envelope exactly—the peak demand $\max(v_x^2|\kappa|)$ over the lap

TABLE III
REFERENCE TIRE MODEL OF THE PLANT, OBTAINED BY FITTING THE MAGIC FORMULA DIRECTLY TO THE SIMULATOR'S PER-WHEEL TELEMETRY OVER A DEDICATED LIMIT-EXCITATION RUN REACHING $\alpha = 0.37$ RAD, AXLE $\mu = 1.05$ AND $1.50\,g$. DIFFERENTIAL EVOLUTION, E CONSTRAINED TO ≥ -2.

Axle	B	C	D	E	RMSE
Front	7.076	1.346	1.009	−2.000	135 N
Rear	7.873	1.383	1.002	−1.024	85 N

TABLE IV
CLOSED-LOOP CONSEQUENCE OF THE DEGENERATE TIRE SET, MEASURED IN A HARNESS DRIVING THE REAL RACELINE FOR 20 S WITH THE PURE-PURSUIT→MPC SWITCH AT $t = 5$ S. LATERAL DEMAND SCALES AS v^2.

Controller model	Max cross-track	Infeasible stages
Plausible prior ($D = 1.5$)	0.03 m	0
Identified set @ 11.1 m/s	0.16 m	0
Identified set @ 15 m/s	0.43 m	0
Identified set @ 27 m/s	41.89 m	29 461

is 12.00 m/s^2, i.e. $1.22\,g$—so the trajectory is feasible *by construction* for the vehicle the optimiser was told about, and only for that vehicle. Measured against the plant's actual steady-state ceiling of $\approx$ 9.8 m/s^2, 11.9 % of the lap is infeasible (Fig. 4c).

Table III is the reference against which the on-track identifier is scored. Note $D \approx 1.0$, not the 1.5 of the wheel material coefficient—the distinction discussed in Section III-C.

B. The failure mode, reproduced

Running the baseline pipeline unmodified on this platform yields

$$\begin{aligned}\text{C\_Pf} &= [5.2874,\ 1.5536,\ \mathbf{0.4},\ \mathbf{1.0}],\\ \text{C\_Pr} &= [19.3236,\ \mathbf{1.2},\ 0.4046,\ -0.1292],\end{aligned}$$

with the bold entries sitting exactly on their box constraints and B_r just under its upper bound: five of eight coefficients on or against a rail. The implied physics is not merely inaccurate but qualitatively wrong—a grip ceiling of 3.95 m/s^2 ($0.40\,g$) and a front axle stiffness of 4012 N/rad, meaning the model believes it needs $\approx$ 0.30 rad of front slip to generate $1\,g$, well beyond the vehicle's 0.2793 rad steering limit. The model essentially believes the car cannot corner.

Table IV shows why this is invisible until it is not.

C. Isolating the cause

Four hypotheses were formulated and tested against measurement rather than argued from code. Table V records the outcome, including the two that were wrong; we report these because the eliminated hypotheses are as informative as the confirmed one.

The decisive experiment is an offline round-trip with the residual network zeroed, so that only the rollout is under test: a synthetic $D = 1.5$ tire, rolled out at 4 m/s and re-fitted, returns $D_f = 0.753$, $D_r = 0.737$ with three coefficients railed; the same tire rolled out at 8–20 m/s returns $D = 1.40$–1.61 with nothing railed. Since the tire is known exactly and the residual model is absent, this isolates the rollout configuration as the cause and confirms (8).

TABLE V
HYPOTHESES TESTED AGAINST THE SIMULATOR'S OWN TELEMETRY.

Hypothesis	Verdict
Friction warm-start measures utilised friction over a low-slip mask	*Structurally confirmed, not the cause.* Utilisation measures 0.044 median, 0.46 peak at 13 m/s. Corrected anyway (Sec. III-J).
Static D prior below its own lower bound	*Confirmed, secondary.* Corrected; multi-starts 1→8.
Only BCD identifiable at low slip	*Confirmed, and it is the mechanism*—but caused by the rollout speed, not by the recorded lap (Sec. III-G).
`/odom` twist in world frame	*Refuted.* corr = 0.93, RMS 0.09 m/s against a body-frame interpretation.
Steady-state force model (7) inadequate	*Refuted as a cause.* corr = 0.993, slope 0.94 against the simulator's own axle force.
δ taken from command, not achieved	*Confirmed, significant.* Ratio 0.76; worth 20 % of front stiffness.

D. Results after correction

With the six defects of Table VI corrected, three consecutive live identifications return no coefficient on a bound. Peak friction coefficients land at $D_f, D_r \in [1.05, 1.18]$ against the reference 1.009/1.002 of Table III; axle cornering stiffnesses land at $C_f \approx$ 21 kN/rad and $C_r \approx$ 20 kN/rad against the simulator's measured 17–19 kN/rad; the sign of the understeer gradient is correct ($l_f C_f < l_r C_r$) on every run, matching the measured plant; and one-step prediction on held-out data gives $R^2 = 0.97$ for v_y and $R^2 = 0.99$ for ω. The peak-friction drift across the six co-identification iterations is removed: where the unbounded sweep walked D from 1.01 to 1.87, the bounded sweep with a frozen prior settles inside $[1.05, 1.18]$.

E. Ablations

Two ablation axes are exposed by the implementation and should be reported in the final version of this paper. Neither has yet been run to completion, and we mark them as such rather than reporting placeholder numbers.

- **Residual architecture**: `baseline` MLP vs `wide` vs `physics_inputs` vs `ensemble` vs `s4` (S4D), scored on one-step and 10-step RMSE and on the identified coefficients. The relevant comparison in the literature is [14], which reports a lateral-force RMSE reduction exceeding 60 % for an S4 residual over conventional architectures; we should expect a smaller effect here because our sequences are short and our residual is dominated by a correctable measurement bias rather than by unmodelled temporal dynamics.
- **Loss mode**: `mse` vs `physics_informed` (dynamics-consistency + symmetry + smoothness terms), on the same dataset, comparing one-step metrics, per-iteration coefficient trajectories, and closed-loop tracking.

F. Negative result: the raceline, not the model

With the identifier corrected, the closed-loop leg of the validation (Section III-L, level 3) does *not* complete, and

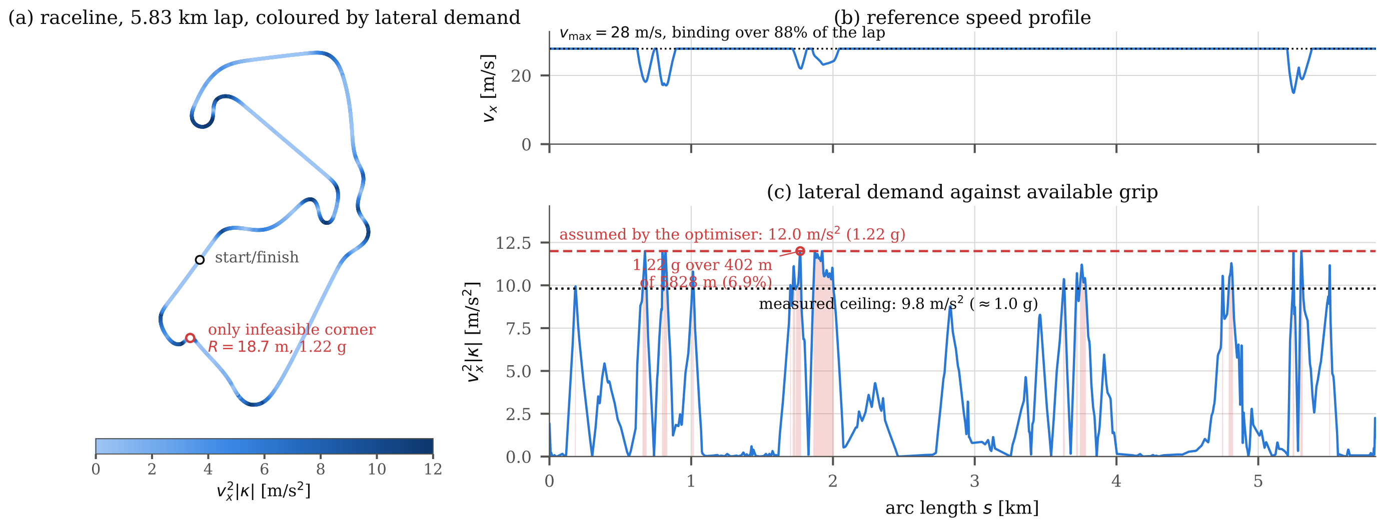

Fig. 4. The optimised raceline used for every experiment in this paper, generated with the TUM `global_racetrajectory_optimization` package [28] under its minimum-curvature formulation [21] from a recorded centreline of the CARLA `silverstone` circuit. (a) Track layout, coloured by the reference speed; the marked corner ($R = 18.7\,\mathrm{m}$) is the one that sets the feasibility limit. (b) Reference speed profile, 14.90–46.02 m/s. (c) Lateral demand $v_x^2|\kappa|$ along the lap. The profile saturates the constant $12.0\,\mathrm{m/s^2}$ envelope the optimiser was given in its `ggv` input (dashed) but not the grip this vehicle actually has (dotted, $\approx 1.0\,g$, measured): the shaded segments, 11.9 % of the lap, demand more than the plant can deliver. This is the trajectory the corrected identifier's admissibility gate rejects, and Section IV-F argues the gate is right to do so.

TABLE VI
DEFECTS IDENTIFIED, CORRECTED AND REGRESSION-TESTED IN THIS WORK. EFFECT COLUMN REPORTS THE MEASURED CHANGE WHERE ONE WAS QUANTIFIED.

Defect	Root cause	Correction	Measured effect
Degenerate fit, D on its bound	Pacejka fit runs on a synthetic rollout whose speed was clipped to 2–4 m/s; $\mu_{\mathrm{reach}} = 0.43$	Rollout speed from measured lap speed, floored at 7 m/s; warning when (8) $< D$	D: 0.40 (railed) $\rightarrow$ 1.05–1.18; C_f: 4.0 $\rightarrow$ 21 kN/rad
Identified set never reached the controller	Trainer wrote coefficients to file but returned nothing; the node re-read a static prior file and submitted *that*	Trainer returns the identified coefficients; node submits them	Service received the prior verbatim while the fit had produced a different set
D drifts upward across iterations	Sweep extrapolated the residual network far past its training range, and each answer became its own regularisation target	Sweep bounded at $1.5 \times P_{99}(\lvert\delta\rvert)$; regularisation target frozen at the configured prior	1.01 $\rightarrow$ 1.87 drift removed; settles 1.05–1.18
Front stiffness underestimated	Slip angle built from the steering *setpoint*	Achieved road-wheel angle from `feedback/steering_angle`, with fallback	C_f: 13725 $\rightarrow$ 18486 N/rad (truth 17198)
Normal loads 10–15% low	m copied from the configuration file (240 kg) against a simulated 269.6 kg; $l_{wb} \neq l_f + l_r$	Measured mass and geometry; $l_{wb} \equiv l_f + l_r$ enforced by test	$F_{z,f}/F_{z,r}$: 1222/1134 $\rightarrow$ 1345.6/1298.9 N
Friction prior pinned near 0.4	Median of *utilised* friction over an explicitly *low-slip* mask	P_{99} over the whole buffer, applied as a floor on the prior	Median 0.044 vs peak 0.46 at 13 m/s

the reason is instructive. The identified model reports a grip ceiling of 1.02–1.16 g; the measured plant sustains $\approx 1.0\,g$ in steady state; the raceline used in this study demands $\max(v_x^2|\kappa|) = 12.00\,\mathrm{m/s^2} = 1.22\,g$ ($P_{99} = 11.90\,\mathrm{m/s^2}$). The admissibility gate of Section III-K therefore rejects every identification—correctly. The trajectory is infeasible for this vehicle, and the corrected identifier is what made that visible.

The cause is not obscure, and it is worth naming because it is a failure mode any model-based stack can inherit. As Section IV-A1 records, the trajectory optimiser was handed a `ggv` envelope asserting a constant $12.0\,\mathrm{m/s^2}$ of lateral capability and a vehicle mass of 204 kg. It did its job: the profile saturates that envelope to the last digit. The envelope was simply not this vehicle's. The identification pipeline and the trajectory were parameterised from two different descriptions of the same car, and nothing in the stack compared them until the gate did.

We deliberately do not loosen the gate. The quantitative consequence for trajectory regeneration is: the tightest corner has $|\kappa| = 0.0536\,/\mathrm{m}$, which at the measured grip permits $\sqrt{g\mu/|\kappa|} = 13.5\,\mathrm{m/s}$ (13.7–14.6 m/s at the identified ceiling of 1.02–1.16 g); capping the speed profile at 13 m/s brings peak demand to 9.05 m/s^2 (0.92 g), whereas 15 m/s returns it to 12.00 m/s^2. Regenerating the raceline with the *measured* vehicle ($m = 269.6$ kg, $\mu_{\mathrm{peak}} = 1.0$, $v_{\max} \approx$ 13–14 m/s) is the prerequisite for the closed-loop experiment, and is the immediate next step (Section VI).

G. Required visuals

The following figures and tables are needed to support the claims above. Data sources are ROS 2 topics or CSV exports already produced by the pipeline; no figure below requires new instrumentation. Generation recipes are collected in `paper/figures/README.md`.

a) Fig. 1 — Identification pipeline: Block diagram. The four stages of Section III-B in a loop, with this paper's changes highlighted: excitation-aware rollout speed, extrapolation-bounded sweep, frozen regularisation target, and the admissibility gate on the output arrow to the MPC.

b) Fig. 2 — Formula AI vehicle in CARLA: Done (`figures/fig2.png`): a front view of `asurt_fsai` on the start/finish straight of the `silverstone` map. Optional improvement: pair it with a top-view schematic annotated with l_f, l_r, L, wheel radius, the 16° steering limit and the IMU/GNSS mount points, so the geometry of Table I is readable off the figure rather than only from the caption.

c) Fig. 3 — CARLA_ASU_BRIDGE *architecture: Done* (`figures/bridge_architecture.tex`, vector Ti*k*Z). Four columns: CARLA server; CARLA_ASU_BRIDGE lifecycle node; ROS 2 topic layer; the autonomy stack. The ground-truth path is drawn reaching only the benchmark nodes, and the gated `mpc/update_params` interface is drawn inside the stack.

d) Fig. 5 — Example telemetry: Stacked line plot, 6 panels sharing a time axis, over 30 s of identification data: (a) v_x [m/s]; (b) v_y and ω (twin axis); (c) commanded vs achieved δ [rad], which visualises the 0.76 ratio; (d) a_x, a_y from the IMU; (e) per-axle F_y from `feedback/tire_forces` against the value recovered by (7); (f) per-axle α. Source: a rosbag of `/odom`, `/drive`, `feedback/steering_angle`, `feedback/imu`, `feedback/tire_forces`.

e) Fig. 6 — Reachable friction: Line plot. μ_{reach} from (8) versus rollout speed $v \in [2, 20]$ m/s, one curve per $\delta_{\mathrm{sweep}} \in \{0.1, 0.2, 0.34, 0.4\}$ rad, with a horizontal line at the plant's measured $\mu = 1.0$, a vertical band at the old 2–4 m/s clip, and a marker at the corrected operating point. Overlay the 1:10 platform's L as dashed curves to show why the defect is scale-latent. This is the paper's central explanatory figure.

f) Fig. 7 — Excitation and identifiability: Two-panel figure. (a) Histograms of α_f and α_r actually entering the fit, before and after the rollout correction, with the reference tire's peak-force slip angle marked. (b) (α, F_y) scatter of the synthetic data with the fitted curve overlaid, before and after; the "before" panel should show all data on the linear ramp.

g) Fig. 8 — Coefficient trajectories: Line plot, 2×4 grid (front/rear $\times$ B, C, D, E), value versus co-identification iteration $1 \ldots 6$, two series (unbounded sweep with tracking prior; bounded sweep with frozen prior), horizontal dashed lines at the box constraints and at the reference values of Table III. This figure carries the drift result.

h) Fig. 9 — Identified tire models: Line plot, two panels (front, rear). F_y versus α over $[0, 0.4]$ rad: the reference fit of Table III, the degenerate baseline result, the corrected result (three runs, shaded band), and the raw (α, F_y) scatter from `feedback/tire_forces` underneath. Directly parallels Fig. 6 of [1] and is the primary accuracy claim.

i) Fig. 10 — One-step prediction: Two-panel time series plus residual histogram. Measured versus predicted v_y and ω over 10 s of held-out data, with the residual distribution inset; annotate RMSE and R^2.

j) Fig. 11 — Regenerated speed profile (pending): Scope narrowed: the demand-versus-grip comparison is now Fig. 4(c), drawn from the real trajectory. What remains is the proposed fix — the profile after regeneration for the measured vehicle, plotted on the same axes so the two read as before and after. A naive speed cap bounds the result (13 m/s gives 9.05 m/s^2, 14 m/s gives 10.50 m/s^2) but understates it, since the optimiser will also reshape the line; produce it by re-running the optimiser with a corrected vehicle configuration, not by clipping the existing profile.

k) Fig. 12 — Cornering stiffness comparison: Grouped bar chart. C_f and C_r for: simulator ground truth, degenerate baseline fit, command-referenced fit, achieved-steering fit, and corrected fit. Annotate the understeer condition $l_f C_f$ vs $l_r C_r$ per group.

l) Fig. 13 — Closed-loop tracking (pending): Two-panel figure, to be produced after raceline regeneration. (a) Cross-track error versus time with the controller-active state shaded; (b) track overlay of the driven path against the raceline. Report lap time, mean and maximum lateral deviation, and steering sign reversals per control step.

m) Tables: Table I (configured vs measured plant), Table II (interface), Table III (reference tire), Table IV (closed-loop consequence), Table V (hypotheses), Table VI (defects and effects) are specified above. Two further tables are required for the final version: a coefficient table (baseline / corrected / reference, per axle, with C_f, C_r, $a_y^{\max}$, v_{crit} derived); and a validation-metric table (RMSE and R^2 for v_y and ω at 1-, 5- and 10-step horizons, plus per-axle force RMSE against `feedback/tire_forces`), with the noise-sweep protocol of [1] as an optional robustness column.

V. Discussion and Limitations

A. What generalises beyond this platform

The reachable-friction bound (8) is kinematic. It involves no tire parameter, no simulator-specific quantity and no property of the residual model, and it therefore applies to any implementation of the virtual-steady-state-data strategy of [1], in simulation or on hardware. Its practical content is a design rule: *the rollout must be configured to demand at least the friction one intends to identify*, and if it cannot, the peak factor D should be treated as unidentified rather than reported. The diagnostic signature—coefficients resting exactly on box constraints—is likewise generic, and we would argue it should be checked and logged by any bounded tire-fitting routine, since it is otherwise indistinguishable in the output from a successful fit.

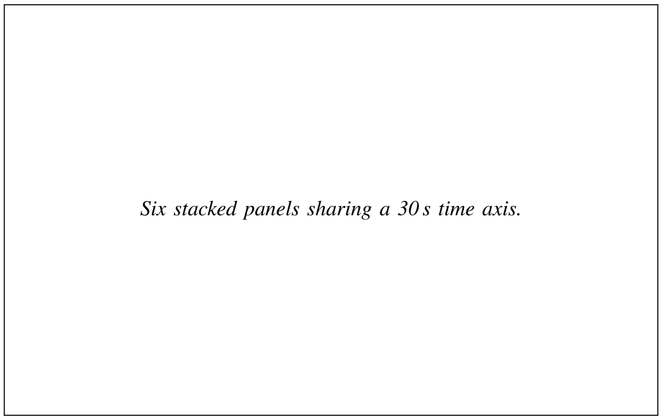

Fig. 5. Example telemetry from one identification run. (a) v_x; (b) v_y and ω; (c) commanded versus achieved road-wheel angle, whose ratio measures 0.76; (d) a_x, a_y; (e) per-axle lateral force from the simulator's wheel telemetry against the value recovered by the steady-state relation (7); (f) per-axle slip angle.

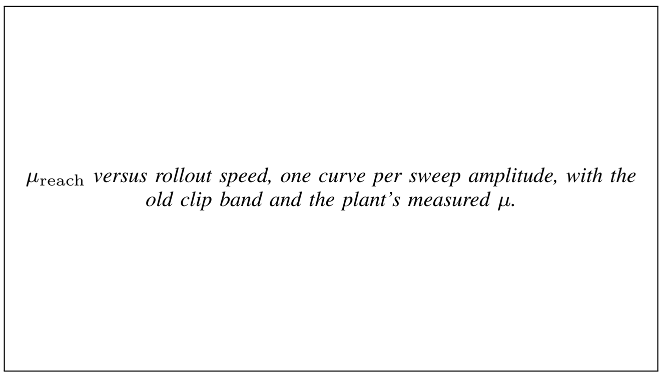

Fig. 6. Reachable friction of the synthetic rollout, $\mu_{\text{reach}} = v^2 \delta_{\text{sweep}}/(gL)$, for the full-scale vehicle ($L = 1.53\,\text{m}$, solid) and a 1:10 platform ($L = 0.32\,\text{m}$, dashed). The shaded band is the rollout speed inherited from the scaled platform; within it the full-scale rollout cannot demand more than $\mu = 0.43$, so the tire's peak factor is not identifiable from it. The same configuration is comfortably above the peak at 1:10, which is why the defect is scale-latent.

The observation that per-coefficient box constraints are insufficient for deployment admissibility is also platform-independent. The two failure modes we encountered—a set that satisfies every box constraint while being degenerate, and a set that satisfies every box constraint while being pairwise oversteering and open-loop unstable within the controller's operating range—are properties of the parameterisation, not of CARLA. Deep Dynamics [13] constrains coefficients to nominal ranges inside the network; we suggest that a second, derived-quantity check belongs at the interface where the model is handed to a controller.

B. What does not generalise

Three results are properties of this specific plant and must not be read as statements about tires.

First, the numerical tire values. The plant is a PhysX wheeled-vehicle model with a configured wheel friction coefficient, not a physically measured tire. Table III is an empirical description of *that* plant's behaviour, and the agreement between our identification and it is evidence that the identifier recovers the plant it observes—not evidence about real tire

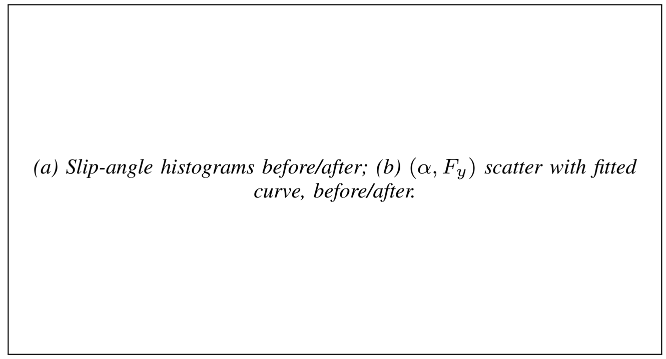

Fig. 7. Excitation entering the Pacejka fit. (a) Distribution of the slip angles present in the synthetic data before and after the rollout correction, with the reference tire's peak-force slip angle marked. (b) The corresponding (α, F_y) scatter with the fitted curve; before the correction all data lies on the initial linear ramp, where only the product BCD is identifiable.

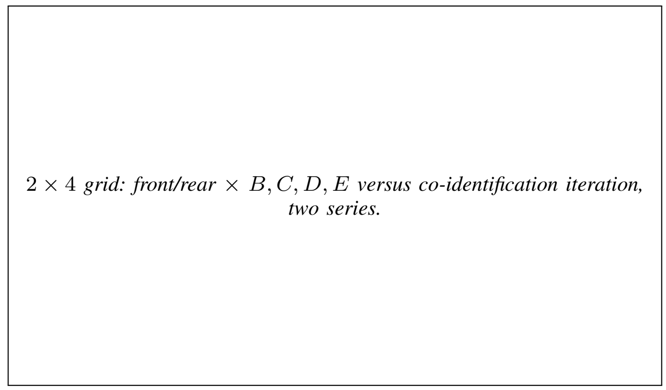

Fig. 8. Coefficient trajectories over the six co-identification iterations, with the unbounded sweep and tracking regularisation target (which walks D from 1.01 to 1.87) against the bounded sweep with a frozen target (which settles in $[1.05, 1.18]$). Dashed lines mark the box constraints and the reference values of Table III.

behaviour. Sagmeister *et al.* [23] show that the sensitivity of controller evaluation to model fidelity is largest precisely near the acceleration limits, which is where this study operates; R-CARLA [22] exists because CARLA's native dynamics are a known weak point for racing-grade work. A hardware replication is therefore necessary before any claim about identification accuracy on a real vehicle.

Second, the gap between the configured wheel friction (1.5) and the realised peak axle friction (1.00–1.05). This is a property of how the PhysX solver composes wheel material friction with load, slip and surface, and provides no information about a physical tire. We report it only because it is a trap: an identification that is tuned to converge on the number written in the simulator's configuration file will be tuned to converge on the wrong number.

Third, the 0.76 achieved/commanded steering ratio arises from CARLA's native Ackermann controller and this vehicle's steering configuration. The *lesson*—identify against the achieved actuator state, not the setpoint—is general and well known in system identification practice; the magnitude is not.

C. Limitations

1) **The closed-loop validation is incomplete.** Level 3 of Section III-L has not been run end-to-end, because the

Two panels (front, rear): $F_y(\alpha)$ for reference, degenerate, and corrected fits over the measured scatter.

Fig. 9. Identified tire models against the plant. Each panel shows the reference fit of Table III, the degenerate result of the untransplanted pipeline, and the corrected result over three consecutive identifications (shaded band), with the raw per-wheel (α, F_y) measurements underneath.

Measured versus predicted v_y and ω on held-out data, with residual histograms inset.

Fig. 10. One-step prediction on a held-out run using the corrected identified model: $R^2 = 0.97$ for lateral velocity and $R^2 = 0.99$ for yaw rate.

corrected identifier's admissibility gate correctly rejects the model for the raceline in hand (Section IV-F). Until a feasible raceline is regenerated, this paper reports open-loop accuracy and closed-loop *consequence* (Table IV), not closed-loop racing performance. We consider stating this preferable to relaxing the gate to produce a result.

2) **No noise sweep.** The baseline's headline robustness claim—a 3.3× RMSE advantage over NLS under injected noise [1]—has not been reproduced on this platform. The odometry noise model exists in the bridge and is disabled for the runs reported here, so the experiment is available; it simply has not been run. Our results are therefore not directly comparable to the baseline's robustness figures.
3) **No NLS baseline on this platform.** We compare against the plant's own telemetry rather than against an NLS identification of the same data. A same-data NLS comparison would strengthen the paper and is straightforward to add.
4) **Single surface, single vehicle, single trajectory.** All results come from one vehicle configuration on one map. The bridge exposes runtime friction override (`control/tire_friction`), so a surface-change adaptation experiment—arguably the strongest argument for on-track identification—is directly supported but not performed here.
5) **Yaw inertia is assumed, not measured.** I_z enters (3) and the stiffness-adaptive sub-stepping directly; see Section A.
6) **Ablations not completed.** The residual-architecture and loss-mode comparisons of Section IV-E are implemented and configurable but not yet run, so no claim is made about the value of the S4D residual or the physics-informed loss on this platform. In particular, we do not claim to reproduce the S4 improvement reported by [14].
7) **Longitudinal dynamics are out of scope.** As in the baseline, only lateral dynamics are identified; combined-slip and load-transfer effects are neglected, and the simulator's longitudinal wheel force channel is drivetrain torque rather than a contact-patch force and is unusable as ground truth for a longitudinal extension without further work.

Pending: demand profile after regenerating the raceline for the measured vehicle, on the same axes as Fig. 4(c).

Fig. 11. Regenerated speed profile. **Not yet produced**: the demand-versus-grip comparison for the *current* trajectory is Fig. 4(c); this figure is its counterpart after the raceline is regenerated for the measured vehicle ($m = 269.6\,\mathrm{kg}$, $\mu_{\mathrm{peak}} = 1.0$), which is the prerequisite for the closed-loop experiment of Fig. 13.

Grouped bars: C_f, C_r across five model sources.

Fig. 12. Axle cornering stiffness $C = F_z B C^{\mathrm{MF}} D$ across model sources: simulator ground truth, the degenerate fit, a fit referenced to the steering command, a fit referenced to the achieved steering angle, and the final corrected fit. The understeer condition $l_f C_f$ versus $l_r C_r$ is annotated per group.

D. Threats to validity

The most serious internal threat is that several of the corrections listed in Table VI were applied together rather than in a factorial design; only the rollout-speed correction was isolated by the zeroed-residual round-trip of Section IV-C. The measured effects attributed to the individual measurement-side corrections come from re-fitting the same recorded run with one change at a time, which controls for the data but not for interaction between the changes. A full ablation over the six defects would settle this and is inexpensive to run offline.

The most serious external threat is the simulator's tire model, discussed above. The mitigation available within this

Pending: cross-track error versus time, and driven path over the raceline.

Fig. 13. Closed-loop tracking under MPC with the identified model. **Not yet produced**: this experiment is blocked on regenerating a feasible raceline for the measured vehicle, see Section IV-F.

study is that every claim is scored against the plant's own per-wheel forces rather than against a secondary model, so the identification result is at least not self-referential.

VI. Conclusion and Future Work

We transplanted a learning-based on-track Pacejka identification pipeline [1] from a 1:10 scaled platform onto a full-scale Formula Student vehicle simulated in CARLA, and found that it fails in a specific, diagnosable and correctable way. The failure is not a solver problem, a bounds problem or a tuning problem: it is an excitation problem created by the pipeline's own virtual-data step, whose reachable friction $\mu_{\text{reach}} = v^2\delta_{\text{sweep}}/(gL)$ depends on the rollout configuration and not on the tire. A rollout speed appropriate at 1:10 caps μ_{reach} at 0.43 at full scale, below the plant's peak, so the peak factor D is not identifiable from the synthetic data and the bounded optimiser resolves the ambiguity on a box edge.

Correcting this—together with an extrapolation bound on the steering sweep, a frozen regularisation target, identification against the achieved rather than commanded road-wheel angle, measured mass and geometry, and a peak-rather-than-median friction floor—produces identifications with no railed coefficient, peak friction coefficients of 1.05–1.18 against a measured 1.00–1.05, axle cornering stiffnesses within 10–20 % of the simulator's own, correct understeer sign, and one-step R^2 of 0.97/0.99. We further argue that per-coefficient box constraints are not a sufficient admissibility criterion for handing a model to a model-based controller, and place gates on derived physical quantities—grip ceiling against trajectory demand, understeer sign and critical speed, and linearisation stiffness—at the interface instead.

The corrected identifier's first useful act was to reject the reference trajectory: the raceline in hand demands 1.22 g from a vehicle that sustains 1.0 g. We report this rather than relaxing the gate.

A. Future work

a) Immediate, to close this paper:

1) Regenerate the raceline for the *measured* vehicle ($m = 269.6\,\text{kg}$, $\mu_{\text{peak}} = 1.0$, $v_{\max} \approx$ 13–14 m/s) and run the closed-loop validation of Fig. 13: lap time, mean and maximum lateral deviation, and steering sign reversals per control step, against a hand-tuned prior model as control.
2) Run the noise sweep of [1] on this platform, using the bridge's odometry noise model, with an NLS identification of the same data as the comparison arm, so that the robustness claim is directly comparable.
3) Complete the residual-architecture and loss-mode ablations of Section IV-E, and a factorial ablation over the six corrections of Table VI.
4) Measure I_z rather than assuming it (Section A).

b) Beyond:

1) **Surface-change adaptation.** The bridge can change tire friction at runtime without respawning, which makes a step-change-in-grip experiment cheap: change μ mid-run and measure the identifier's convergence time and the controller's transient. This is the scenario on-track identification exists for, and it is difficult to stage on hardware.
2) **Excitation-aware data collection.** Equation (8) inverts into a requirement on the manoeuvre. A controller that deliberately inserts bounded slalom or ramp-steer excitation when the identifier reports insufficient slip-angle coverage would close the loop between what the identifier needs and what the vehicle does—moving from passive on-track identification toward active, safety-bounded on-track experiment design.
3) **Uncertainty-aware output.** The gates of Section III-K are currently binary. Reporting a posterior over the coefficients—the Bayesian variational identification path already implemented alongside the deterministic solvers—would let the controller weight the model by its confidence rather than accept or reject it.
4) **Hardware replication.** The decisive test of every claim in Section V is a physical Formula Student vehicle. The identification code is unchanged between simulation and hardware; what changes is the availability of ground truth, which is exactly what makes the simulation study a necessary precursor rather than a substitute.

Appendix A
Assumptions and Reproducibility Notes

The following are assumptions or design choices rather than measured facts. They are listed explicitly so that a reader can judge how far each result depends on them, and so that each can be discharged by a specific measurement.

A1. Yaw inertia I_z

$I_z = 51.1\,\text{kg}\,\text{m}^2$ is carried in the model file and is *not* measured. It enters (3) directly and therefore scales the yaw-rate residual the network is trained on, and it enters the stiffness-adaptive sub-stepping of Section III-K through $l_f^2 C_f + l_r^2 C_r$ over I_z. *How to discharge:* apply a known yaw moment (a step steer at constant speed with the resulting axle forces read from `feedback/tire_forces`) and fit I_z from the measured $\dot{\omega}$; or read the value the physics engine is using directly from the actor's inertia tensor.

A2. Centre-of-gravity split

$l_f = 0.738142\,\mathrm{m}$, $l_r = 0.795362\,\mathrm{m}$ are taken from the controller configuration and confirmed by the vehicle owner. The measured static load split (51.2/48.8 front/rear) implies a marginally different split than the geometry does (51.9/48.1) — a $1.4\,\%$ disagreement. We have not resolved it, and it is below the level at which any conclusion in this paper would change.

A3. CG height and load transfer

$h_{\mathrm{cg}} = 0.3\,\mathrm{m}$ is carried in the model file and unused: the single-track model of Section III-A neglects load transfer, following the baseline. Normal loads are static. On a full-scale vehicle at $1\,g$ this is a stronger approximation than at 1:10, and it is a candidate explanation for part of the residual the network absorbs.

A4. Steering-angle sign and units

`feedback/steering_angle` is published in degrees and is treated as the front road-wheel angle, positive left, matching the ROS body frame. It was verified to agree with the front-left wheel joint state to within 0.5 %. Left and right road wheels are not distinguished; the single-track δ is taken as this single reported value.

A5. Steady-state force recovery

Equation (7) is not a force measurement — it is fully determined by $v_x\omega$ — so it misattributes transients. It is applied only to the *synthetic*, quasi-steady rollout, where the steady-state assumption is constructed rather than hoped for. Its adequacy was nonetheless checked on measured data against the simulator's own axle force (correlation 0.993, slope 0.94); we treat that as validating the relation, not as validating a steady-state assumption on recorded laps.

A6. The residual network is not the plant

The corrected model $\hat{\mathbf{x}}_{k+1} + e_k$ is queried by the rollout at inputs the network has only partially seen. The extrapolation bound of Section III-H limits but does not eliminate this: the sweep still reaches $1.5\times$ the observed 99th percentile of $|\delta|$. The choice of 1.5 is empirical, selected because it was sufficient to reach the tire's peak at the corrected rollout speed while removing the observed drift; it is not derived.

A7. Simulator determinism and timing

CARLA runs in synchronous mode with a 0.033 s physics step, but sim time advances faster than wall time (observed $\approx 1.9\times$ on our hardware). All identification data is stamped from the simulation clock. We assume the physics step is uniform and that dead-reckoned odometry samples (disabled here via `follow_server_rate`) do not enter the identification buffer.

A8. Software configuration of record

Identification and control workspace: ROS 2 Humble, Python 3.10 (required — ROS 2 Humble ships CPython 3.10 ABI extensions only), PyTorch with CUDA on an RTX 2080-class GPU. The bridge is built and sourced as a separate overlay on top of that workspace, so that its message packages (`sim_manager_msgs`) resolve for the ground-truth topics; CARLA server 0.9.16 / Unreal Engine 4.26; `acados` 0.5.5 with partial-condensing HPIPM. Identification runs at 50 Hz with a 30 s buffer, six co-identification iterations, 200 training epochs, full-batch Adam at 5×10^{-4}.

A9. Regression tests

Eight tests pin the identifiability signature (no fitted coefficient on a bound; $l_{wb} \equiv l_f + l_r$; the rollout speed derivation) and run under `colcon test`, so that the degenerate-fit signature cannot return silently. This is a reproducibility mechanism rather than a scientific result, but it is the mechanism that would catch a regression of the paper's central finding.

APPENDIX B
REFERENCE IMPLEMENTATION

The identification package, the ROS 2 bridge and the controller stack used in this paper are available as `On-Track-SysID` (extended from [1]), CARLA_ASU_BRIDGE [29], and the accompanying MPC and benchmarking packages. The benchmarking packages compute one-step and multi-step prediction metrics online (RMSE, MAE, NRMSE, MaxAE, bias, standard deviation and R^2 by Welford's algorithm), compare estimated against ground-truth per-wheel lateral forces, and record controller-switching behaviour, and were built for this study.

ACKNOWLEDGMENT

The ON-TRACK-SYSID pipeline studied here originates from the ForzaETH Race Stack and the work of Dikici *et al.* [1], released under an open-source licence; this paper reports a port, a correction, and an extension of that work rather than a reimplementation from scratch.

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